



RV College of Engineering®

Mysore Road, RV Vidyaniketan Post, Bengaluru - 560059, Karnataka, India

EXPERIMENTAL INVESTIGATION OF LOW-COST JET SPRAY BREAKUP FOR AERO PROPULSIVE APPLICATIONS

PROJECT BASED LEARNING REPORT(PBL) COURSE: MECHANICS OF FLUIDS

submitted by

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under the guidance of

Dr. Karthik Vel E Assistant Professor Dept of Aerospace Engg	Dr. Balaguru Pandian Assistant professor Dept of Aerospace Engg	Dr. R. Supreeth Associate professor & HoD Dept of Aerospace Engg
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in partial fulfilment for the award of degree of

Bachelor of Engineering

in

Department of AEROSPACE ENGINEERING

2025-2026



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DEPARTMENT OF AEROSPACE ENGINEERING

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CERTIFICATE

Certified that the Project based learning(PBL) titled 'Experimental Investigation of Low Cost Jet Spray Breakup for Aero Propulsive Applications' of Mechanics of Fluids Course is carried out by Treya Moodithaya (1RV24AS057), R Tushti (1RV24AS038), Lakshmi D L (1RV24AS026) and Yashaswi Chandrashekar (1RV24AS064) who are bonafide students of RV College of Engineering, Bengaluru, in partial fulfillment for the award of Degree of Bachelor of Engineering, RVCE affiliated to Visvesvaraya Technological University, Belagavi during the year 2025-2026. It is certified that all corrections/suggestions indicated for the internal assessment have been incorporated in the report deposited in the department library.

The project report has been approved as it satisfies the academic requirements in respect of project work prescribed by the institution for the said Degree.

Project Guide

Head of Department

External Viva Examination

Name of Examiner

Signature with Date

1.

2.



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DECLARATION

We the students of the Third semester B.E., Department of **Aerospace Engineering**, RV College of Engineering, Bengaluru-560059, hereby declare that the project titled **‘Experimental Investigation of Low-Cost Jet Spray Breakup’** has been carried out by us and submitted in partial fulfilment of the program requirements for the award of Degree in Bachelor of Engineering, RVCE of the **Visvesvaraya Technological University, Belagavi** during the year **2025-2026**.

Further, we declare that the content of the dissertation has not been submitted previously by anybody for the award of any Degree or Diploma to any other University.

We also declare that any Intellectual property rights generated out of this project carried out at RVCE will be the property of RV College of Engineering, Bengaluru and we will be among the authors of the same.

Place: Bangalore

Date:

Name	USN	Department	Signature
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1) Introduction

In high-performance propulsive systems, such as Gas Turbines or Liquid Rocket Engines (LREs), the conversion of chemical energy into thrust is limited by the mixing rate of the propellant. Since fuel is injected as a liquid, it must undergo atomization—the process of breaking a bulk liquid into a fine spray of droplets—before it can evaporate and react with the oxidizer.

The efficiency of the engine is directly proportional to the total surface area of the fuel spray. A finer spray increases the heat transfer rate, leading to faster evaporation and a more compact combustion zone. This research utilizes a high-pressure Bosch nozzle as a surrogate for industrial injectors to analyse the fundamental transition from a coherent liquid jet to a fully developed spray. The optimization of aero propulsive combustion relies on precise control over fuel-air mixing, yet quantifying the spray morphology requires complex equipment. This investigation utilizes a low-cost high-pressure water system to experimentally derive critical atomization metric such as break up length and droplet distribution. The study aims to bridge the gap between accessible experimental hardware and rigorous data requirements needed to predict combustion efficiency and thrust performance.

1.1) Problem Definition:

The optimization of aero propulsive combustion relies on precise control over fuel-air mixing, yet quantifying the spray morphology requires complex equipment. This investigation utilizes a low-cost high-pressure water system to experimentally derive critical atomization metric such as break up length and droplet distribution. The study aims to bridge the gap between accessible experimental hardware and rigorous data requirements needed to predict combustion efficiency and thrust performance.

1.2) Objectives:

The primary aim of this study is to characterize the morphological evolution of a high-pressure jet.

- Identify the primary and secondary breakup regimes as a function of injection pressure.

- Quantify the Breakup Length and Spray Cone Angle to assess spatial distribution.
- Determine the Sauter Mean Diameter (SMD) to predict the evaporation potential of the spray in a thermal environment.

1.3) Theory:

The disintegration of a liquid jet is not instantaneous; it is a multi-stage fluid dynamic event governed by hydrodynamic instabilities.

The primary breakup regime: As the liquid exits the Bosch nozzle, the interface between the high-velocity liquid and the relatively stagnant ambient air becomes unstable. This is primarily driven by Kelvin-Helmholtz (KH) Instability. Small perturbations on the jet surface are amplified by aerodynamic forces until they form "ligaments."

The secondary breakup regime: Once droplets are formed, they undergo further fragmentation. The nature of this breakup depends on the Weber Number, which compares the disruptive aerodynamic drag to the restorative surface tension.

1.4) Key Characterization Parameters:

I. Breakup Length The distance from the nozzle orifice to the point where the continuous liquid core fragments. In propulsion, a shorter breakup length is critical for "quick-start" ignition and reducing the physical length of the combustor.

II. Spray Cone Angle: This defines the volumetric spread of the fuel. A wider angle ensures better air-fuel mixing (macro-mixing) but must be balanced to prevent fuel from impinging on the chamber walls, which causes structural cooling issues.

III. Sauter Mean Diameter: Standard arithmetic means are insufficient for spray analysis because evaporation is a surface-area-dependent process. SMD is defined as:

$$D_{32} = \frac{\sum n_i d_i^3}{\sum n_i d_i^2}$$

where n is the number of droplets of diameter d. This value represents a drop that has the same volume-to-surface area ratio as the entire spray, making it the gold standard for calculating combustion efficiency.

1.5) Literature Review:

Paper	Summary of Content
Experimental and numerical investigations on the spray characteristics of liquid-gas pintle injector <i>Wenyuan Zhou, Xu Xu, et al. (2022)</i>	This study examines the spray performance of liquid-gas pintle injectors using water and nitrogen as simulants. Researchers utilized high-speed photography and Malvern measuring instruments to analyse spray patterns, spray angles, and Sauter mean diameter (SMD). A key finding is the significant role of the Total Momentum Ratio (TMR) in determining the spray angle and droplet size. The study integrates experimental data with a VOF-LPT coupling numerical algorithm for comprehensive analysis. Findings provide a theoretical basis for optimizing pintle injectors in rocket engines.
Numerical simulation and experimental study of jet breakup using a water dispersal needle in irrigation sprinklers <i>Xuwei Pan, Yue Jiang, et al. (2024)</i>	The paper investigates how a water dispersal needle affects jet breakup and water distribution uniformity in irrigation sprinklers. Using the Volume-of-Fluid (VOF) method and SST k- ω model, the study simulates the impact of needle insertion depth, cone angle, and distance from the nozzle. Results indicate that increasing needle depth enhances jet breakup but can reduce the throw radius. Experimental validation confirms that specific needle geometries optimize air entrainment and droplet formation. The research aims to improve irrigation efficiency through better spray uniformity.
Breakup of a laminar liquid jet by coaxial non-swirling and swirling air streams	This research explores the primary breakup of a laminar liquid jet subjected to coaxial air streams with varying degrees of swirl. The study identifies distinct breakup regimes such as fiber-like and membrane-type breakup, depending on the Weber number and

Paper	Summary of Content
<i>Yifan Liang, Lars Christian Johansen, and Mark Linne (2022)</i>	swirl intensity. It highlights that introducing a swirl component to the air stream significantly accelerates the jet disintegration process. High-speed imaging and statistical analysis (using Hamiltonian Monte Carlo methods) were employed to characterize the spray morphology. The findings are critical for improving fuel-air mixing in combustion systems.
Effect of atomizer and cross-flows on liquid and gelled Jet A-1 fuel injection <i>Balaguru Pandian and Rajarshi Das (2023)</i>	This study compares the atomization characteristics of liquid and gelled Jet A-1 fuels using a modified internally impinging air-blast atomizer. The authors investigate the impact of cross-flow velocities and injection pressures on spray development and droplet size. Results show that gelled fuels, due to their non-Newtonian behaviour and higher apparent viscosity, produce larger droplets and require higher momentum for effective breakup compared to liquid fuel. Qualitative flow visualization and quantitative drop size measurements were used to map the fuel spray distance in transverse and axial directions. The work provides insights into the feasibility of using gelled propellants in propulsion systems.

Table 1.1: Literature Review

2) Methodology:

- A Bosch AQT-3512 pressure washer was used to analyze the jet spray breakup coming from nozzles of 3 different widths: 3mm, 4mm and 5 mm respectively.
- The mass flow rate from each nozzle was recorded by measuring the weight of water collected over a period of 20 seconds of operation.

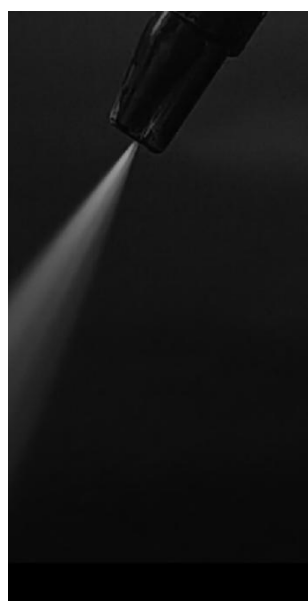
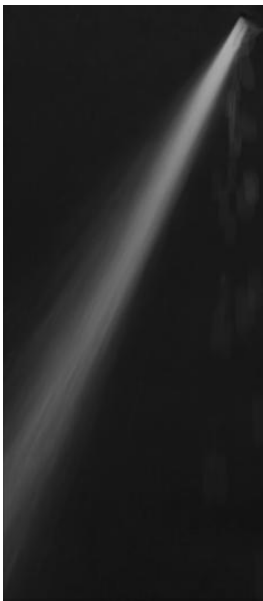
The detailed experimental setup is given below

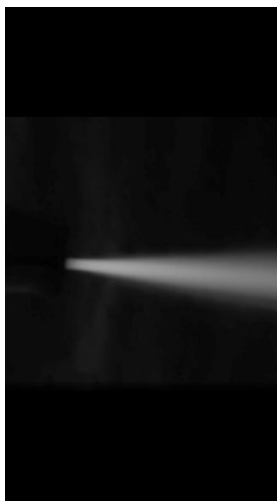


Figure 2.1: Experimental Setup

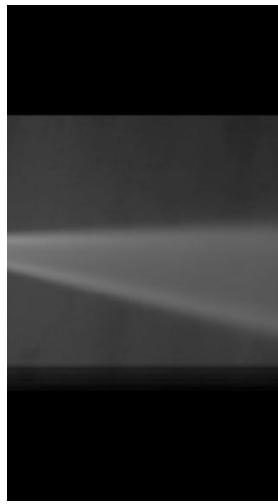
3) Observations:

Image captured:

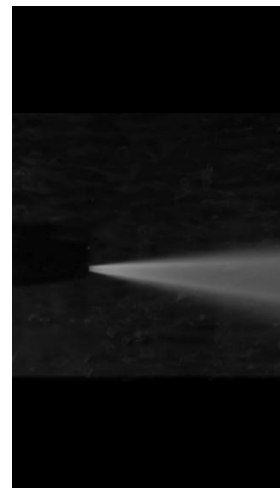




Trial 1:
Mass flow rate =
0.105 kg/s
Nozzle width = 3
mm



Trial 2:
Mass flow rate =
0.145 kg/s
Nozzle width = 4 mm



Trial 3:
Mass flow rate =
0.145 kg/s
Nozzle width = 5 mm

Figure 3.1: Image Observations

4) MATLAB Code for Image Processing:

4.1) Code to calculate the BL, PL:

```
%% 1. LOAD & CALIBRATE
[file, path] = uigetfile({'*.png;*.jpg;*.tif'}, 'Select Averaged Image');
img = imread(fullfile(path, file));
if size(img,3)==3, img=rgb2gray(img); end

% Calibration
imshow(img); title('Click LEFT edge of nozzle, then RIGHT edge');
[kx, ky] = ginput(2);
pxWidth = abs(kx(2) - kx(1));
realSize = input('Enter nozzle diameter in mm: ');
scale = realSize / pxWidth;

title('Crop the spray area tightly and double-click');
rect = getrect;
img = imcrop(img, rect);
close all;

%% 2.BREAKUP & PENETRATION
vProfile = smooth(double(sum(img, 2)));
vProfile = vProfile - min(vProfile);

startRow = find(vProfile > max(vProfile)*0.1, 1, 'first');

% BREAKUP LENGTH
[maxVal, maxIdx] = max(vProfile);
breakupRow = find(vProfile(maxIdx:end) < maxVal*0.6, 1, 'first') + maxIdx;

% PENETRATION LENGTH
penRow = find(vProfile(maxIdx:end) < maxVal*0.05, 1, 'first') + maxIdx;

% Convert to mm
```

```

breakupLengthMM = (breakupRow - startRow) * scale;
penetrationMM = (penRow - startRow) * scale;

%% 3. RESULTS
figure;
plot(vProfile, 'LineWidth', 2); hold on;
xline(breakupRow, '--r', 'Breakup');
xline(penRow, '--g', 'Penetration');
xlabel('Distance from Nozzle (pixels)'); ylabel('Intensity (A.U.)');
title('Spray Axial Intensity Profile');

fprintf('\n--- RESULTS ---\n');
fprintf('Breakup Length: %.2f mm\n', breakupLengthMM);
fprintf('Penetration Length: %.2f mm\n', penetrationMM);

```

4.2) Code to calculate density profile

```

img = imread('trial 1 avg.png');
if size(img,3)==3, img=rgb2gray(img); end

centerCol = round(size(img,2)/2);
densityProfile = img(:, centerCol);

% Plotting
figure;
plot(densityProfile, 'LineWidth', 2, 'Color', 'b');
xlabel('Distance from Nozzle (pixels)');
ylabel('Spray Intensity (Density)');
title('Spray Density Distribution (Time-Averaged)');
grid on;

```

4.3) Average and grayscale trials

```

videoPath = fullfile(getenv('TUSHTI R'), 'Downloads', 'trial 3.mp4');
v = VideoReader(videoPath);

%% 2.Stabilization Parameters
numFrames = 11;

anchorFrame = rgb2gray(readFrame(v));
[optimizer, metric] = imregconfig('monomodal');

sumFrame = double(anchorFrame);

fprintf('Starting stabilization for %d frames...\n', numFrames);

%% 3. The Stabilization & Averaging Loop
for i = 2:numFrames
    if hasFrame(v)
        movingFrame = rgb2gray(readFrame(v));

        stabilizedFrame = imregister(movingFrame, anchorFrame, 'translation', optimizer, metric);

        sumFrame = sumFrame + double(stabilizedFrame);

        fprintf('Processed frame %d of %d\n', i, numFrames);
    end
end

%% 4. Final Averaging
averagedImage = uint8(sumFrame / numFrames);

%% 5. Display Result
imshow(averagedImage);
title(['Stabilized Average of ', num2str(numFrames), ' Frames']);

```

```
imwrite(averagedImage, 'stabilized_spray_average.png');
disp('Process complete. Image saved as stabilized_spray_average.png');
```

4.4) Sauter Mean Diameter Calculation

```
clear all; close all; clc;
```

```
%% Video File
```

```
[fileName, filePath] = uigetfile('*.mp4', 'Select the spray video file');
```

```
if isequal(fileName, 0)
    disp('User canceled file selection');
    return;
else
```

```
    videoPath = fullfile(filePath, fileName);
    fprintf('Selected file: %s\n', videoPath);
end
```

```
v = VideoReader(videoPath);
```

```
%% Nozzle Width
```

```
fprintf('\n=== NOZZLE CALIBRATION ===\n');
nozzleDiameter_mm = input('Enter nozzle diameter in mm (e.g., 3, 4, or 5): ');
nozzleDiameter_um = nozzleDiameter_mm * 1000;
```

```
fprintf('\nScanning video to find frame with most droplets...\n');
fprintf('Video has %d frames at %.2f fps\n', v.NumFrames, v.FrameRate);
sampleInterval = max(1, floor(v.NumFrames / 50));
sampleFrames = 1:sampleInterval:v.NumFrames;
```

```
if length(sampleFrames) > 50
    sampleFrames = sampleFrames(1:50);
end
```

```
maxDropletCount = 0;
bestFrameNum = 1;
bestROI = [];
```

```
v.CurrentTime = 0;
firstFrame = readFrame(v);
grayFirstFrame = rgb2gray(firstFrame);
```

```
fprintf('\nSelect the spray region to analyze (will be used for all frames)...\n');
figure('Name', 'Select ROI');
imshow(grayFirstFrame);
title('Draw a rectangle around the spray region to analyze');
```

```
roi = drawrectangle('Color', 'r', 'LineWidth', 2);
wait(roi);
```

```
roiPosition = round(roi.Position);
close(gcf);
```

```
fprintf('\nAnalyzing frames...\n');
for i = 1:length(sampleFrames)
    frameNum = sampleFrames(i);
```

```
    v.CurrentTime = (frameNum - 1) / v.FrameRate;
    frame = readFrame(v);
    grayFrame = rgb2gray(frame);
```

```
    roiFrame = imcrop(grayFrame, roiPosition);
```

```

roiEnhanced = imadjust(roiFrame);
se = strel('disk', 15);
background = imopen(roiEnhanced, se);
roiSubtracted = imsubtract(roiEnhanced, background);
threshold = graythresh(roiSubtracted) * 1.2;
binaryImage = imbinarize(roiSubtracted, threshold);
binaryImage = bwareaopen(binaryImage, 5);
binaryImage = imfill(binaryImage, 'holes');

labeledImage = bwlabel(binaryImage);
numDroplets = max(labeledImage(:));

if numDroplets > maxDropletCount
    maxDropletCount = numDroplets;
    bestFrameNum = frameNum;
end

if mod(i, 10) == 0
    fprintf('  Processed %d/%d sample frames...\n', i, length(sampleFrames));
end
end

fprintf('\nBest frame found: Frame #%d with ~%d droplets\n', bestFrameNum, maxDropletCount);

v.CurrentTime = (bestFrameNum - 1) / v.FrameRate;
selectedFrame = readFrame(v);
grayFrame = rgb2gray(selectedFrame);
roiFrame = imcrop(grayFrame, roiPosition);

fprintf('\nDetecting nozzle for calibration...\n');
figure('Name', 'Nozzle Detection for Calibration');
imshow(grayFrame);
title('Draw a line across the nozzle diameter');
h = drawline('Color', 'y', 'LineWidth', 2);
wait(h);

linePos = h.Position;
nozzlePixels = sqrt((linePos(2,1) - linePos(1,1))^2 + (linePos(2,2) - linePos(1,2))^2);
close(gcf);

scalingFactor = nozzleDiameter_um / nozzlePixels;
fprintf('Nozzle width: %.2f pixels\n', nozzlePixels);
fprintf('Scaling factor: %.2f μm/pixel\n', scalingFactor);

fprintf('\nProcessing best frame for droplet detection...\n');

roiEnhanced = imadjust(roiFrame);

se = strel('disk', 15);
background = imopen(roiEnhanced, se);
roiSubtracted = imsubtract(roiEnhanced, background);

threshold = graythresh(roiSubtracted) * 1.2;
binaryImage = imbinarize(roiSubtracted, threshold);

binaryImage = bwareaopen(binaryImage, 5);
binaryImage = imfill(binaryImage, 'holes');
binaryImage = imerode(binaryImage, strel('disk', 1));

fprintf('Detecting and measuring droplets...\n');

labeledImage = bwlabel(binaryImage);

```

```

measurements = regionprops(labeledImage, 'Area', 'Centroid', 'EquivDiameter', ...
    'MajorAxisLength', 'MinorAxisLength', 'Eccentricity');

numDroplets = length(measurements);
fprintf('Detected %d droplets\n', numDroplets);

if numDroplets == 0
    error('No droplets detected! Try adjusting the threshold or ROI.');
```

end

```

areas = [measurements.Area];
diameters = [measurements.EquivDiameter];

minArea = 5;
maxArea = (nozzlePixels * 0.3)^2 * pi;

validIdx = (areas >= minArea) & (areas <= maxArea);
measurements = measurements(validIdx);
numValidDroplets = sum(validIdx);

fprintf('After filtering: %d valid droplets\n', numValidDroplets);

if numValidDroplets == 0
    error('No valid droplets after filtering! Adjust minArea and maxArea parameters.');
```

end

```

diameters = [measurements.EquivDiameter];
areas = [measurements.Area];

D2 = diameters .^ 2;
D3 = diameters .^ 3;

SMD_pixels = sum(D3) / sum(D2);
SMD_micrometers = SMD_pixels * scalingFactor;

D10 = mean(diameters) * scalingFactor;
D20 = sqrt(mean(D2)) * scalingFactor;
D30 = (mean(D3))^(1/3) * scalingFactor;

fprintf('\n=== DROPLET SIZE STATISTICS ===\n');
fprintf('Frame analyzed: %d (best frame)\n', bestFrameNum);
fprintf('Number of droplets analyzed: %d\n', numValidDroplets);
fprintf('Nozzle diameter: %.1f mm (%.0f μm)\n', nozzleDiameter_mm, nozzleDiameter_um);
fprintf('Calibration: %.2f μm/pixel\n', scalingFactor);
fprintf('D10 (Arithmetic mean): %.2f μm\n', D10);
fprintf('D20 (Surface mean): %.2f μm\n', D20);
fprintf('D30 (Volume mean): %.2f μm\n', D30);
fprintf('SMD/D32 (Sauter mean): %.2f μm\n', SMD_micrometers);
fprintf('Min diameter: %.2f μm\n', min(diameters) * scalingFactor);
fprintf('Max diameter: %.2f μm\n', max(diameters) * scalingFactor);
fprintf('Std deviation: %.2f μm\n', std(diameters) * scalingFactor);

figure('Name', 'Droplet Analysis Results', 'Position', [50 50 1400 800]);

subplot(2,3,1);
imshow(roiFrame);
title(sprintf('Frame #%d - Original ROI', bestFrameNum));

subplot(2,3,2);
imshow(roiSubtracted);
title('Background Subtracted');
```

```

subplot(2,3,3);
imshow(binaryImage);
title(sprintf('Binary Image (%d droplets)', numValidDroplets));

subplot(2,3,4);
imshow(label2rgb(labeledImage, 'jet', 'k', 'shuffle'));
title('Detected Droplets (Color Coded)');

subplot(2,3,5);
imshow(roiFrame);
hold on;
boundaries = bwboundaries(binaryImage);
for k = 1:length(boundaries)
    boundary = boundaries{k};
    plot(boundary(:,2), boundary(:,1), 'g', 'LineWidth', 1);
end
title('Droplet Boundaries');
hold off;

subplot(2,3,6);
histogram(diameters * scalingFactor, 30, 'FaceColor', 'b', 'EdgeColor', 'k');
hold on;
xline(SMD_micrometers, 'r--', 'LineWidth', 2.5, 'Label', sprintf('SMD=%.1fµm', SMD_micrometers));
xline(D10, 'g--', 'LineWidth', 1.5, 'Label', sprintf('D10=%.1fµm', D10));
xlabel('Droplet Diameter (µm)');
ylabel('Frequency');
title('Droplet Size Distribution');
grid on;
legend('Droplets', 'SMD (D32)', 'D10');
hold off;

figure('Name', 'Summary', 'Position', [100 100 800 600]);
imshow(selectedFrame);
hold on;

rectangle('Position', roiPosition, 'EdgeColor', 'r', 'LineWidth', 2);

textStr = sprintf('Frame: %d\nNozzle: %.1f mm\nDroplets: %d\nSMD: %.1f µm', ...
    bestFrameNum, nozzleDiameter_mm, numValidDroplets, SMD_micrometers);
text(20, 40, textStr, 'Color', 'yellow', 'FontSize', 12, 'FontWeight', 'bold', ...
    'BackgroundColor', 'black');

title(sprintf('Best Frame Analysis - %s', fileName), 'Interpreter', 'none');
hold off;

outputFolder = 'Droplet_Analysis_Results';
if ~exist(outputFolder, 'dir')
    mkdir(outputFolder);
end

saveas(gcf, fullfile(outputFolder, sprintf('%s_Summary.png', fileName(1:end-4))));
saveas(figure(2), fullfile(outputFolder, sprintf('%s_Detailed_Analysis.png', fileName(1:end-4))));

resultsFile = fullfile(outputFolder, sprintf('%s_SMD_Results.txt', fileName(1:end-4)));
fid = fopen(resultsFile, 'w');
fprintf(fid, 'Spray Droplet Analysis Results\n');
fprintf(fid, '=====\n\n');
fprintf(fid, 'Video: %s\n', fileName);
fprintf(fid, 'Best Frame: %d (of %d total)\n', bestFrameNum, v.NumFrames);
fprintf(fid, 'Droplets analyzed: %d\n\n', numValidDroplets);

```



```

fprintf(fid, 'CALIBRATION:\n');
fprintf(fid, 'Nozzle diameter: %.1f mm (%.0f  $\mu$ m)\n', nozzleDiameter_mm, nozzleDiameter_um);
fprintf(fid, 'Nozzle width in image: %.2f pixels\n', nozzlePixels);
fprintf(fid, 'Scaling factor: %.2f  $\mu$ m/pixel\n\n', scalingFactor);
fprintf(fid, 'RESULTS:\n');
fprintf(fid, 'D10 (Arithmetic mean): %.2f  $\mu$ m\n', D10);
fprintf(fid, 'D20 (Surface mean): %.2f  $\mu$ m\n', D20);
fprintf(fid, 'D30 (Volume mean): %.2f  $\mu$ m\n', D30);
fprintf(fid, 'SMD/D32 (Sauter mean): %.2f  $\mu$ m\n', SMD_micrometers);
fprintf(fid, 'Min diameter: %.2f  $\mu$ m\n', min(diameters) * scalingFactor);
fprintf(fid, 'Max diameter: %.2f  $\mu$ m\n', max(diameters) * scalingFactor);
fprintf(fid, 'Std deviation: %.2f  $\mu$ m\n', std(diameters) * scalingFactor);
fclose(fid);

csvFile = fullfile(outputFolder, sprintf('%s_Droplet_Data.csv', fileName(1:end-4)));
dropletData = table((diameters * scalingFactor)', areas', ...
    'VariableNames', {'Diameter_um', 'Area_pixels'});
writetable(dropletData, csvFile);

fprintf('\nResults saved to: %s\n', outputFolder);
fprintf('Analysis complete!\n');

```

5) Results and Discussions

5.1) TRIAL 1:



Density distribution:

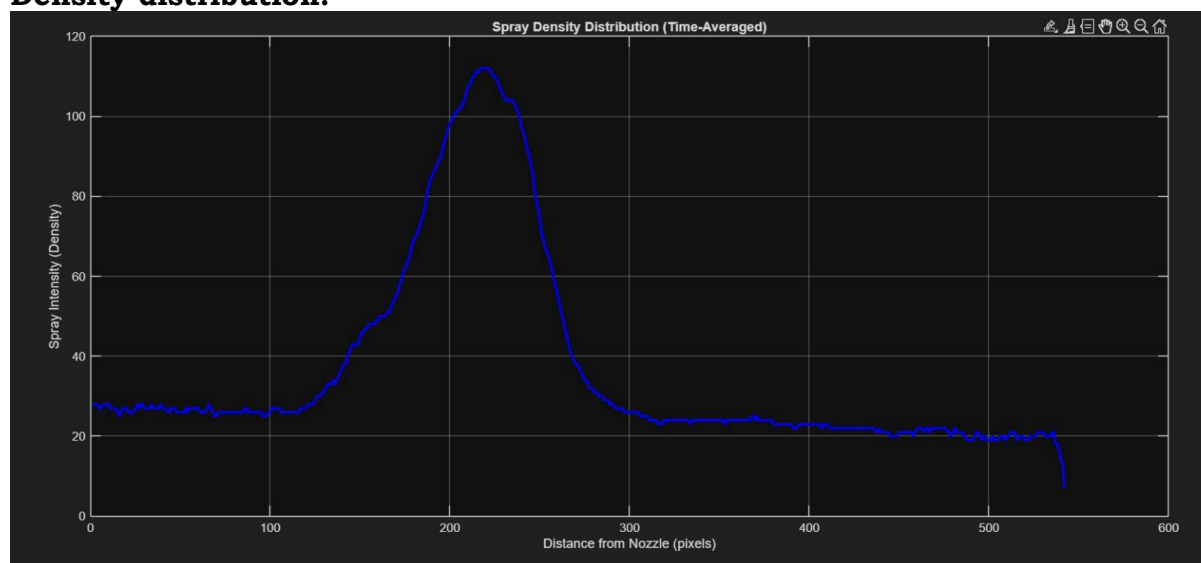


Figure 5.1.1) Density distribution

Breakup Length: 180.19 mm

Penetration Length: 227.71 mm

Intensity distribution:

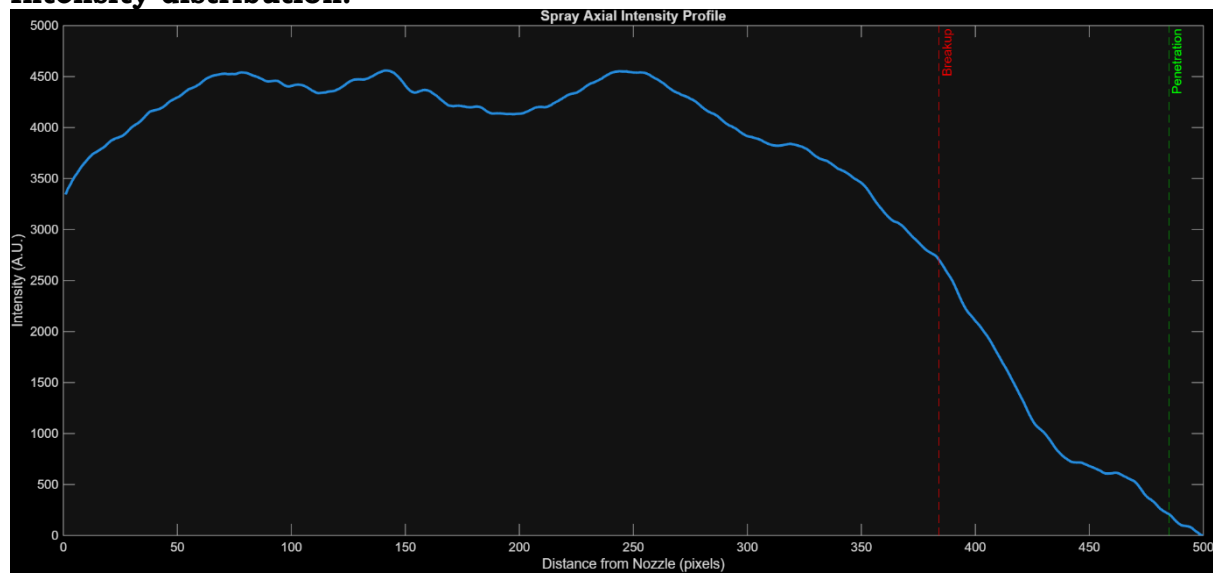


Figure 5.1.2) Intensity Distribution

Spray angle on SolidWorks:

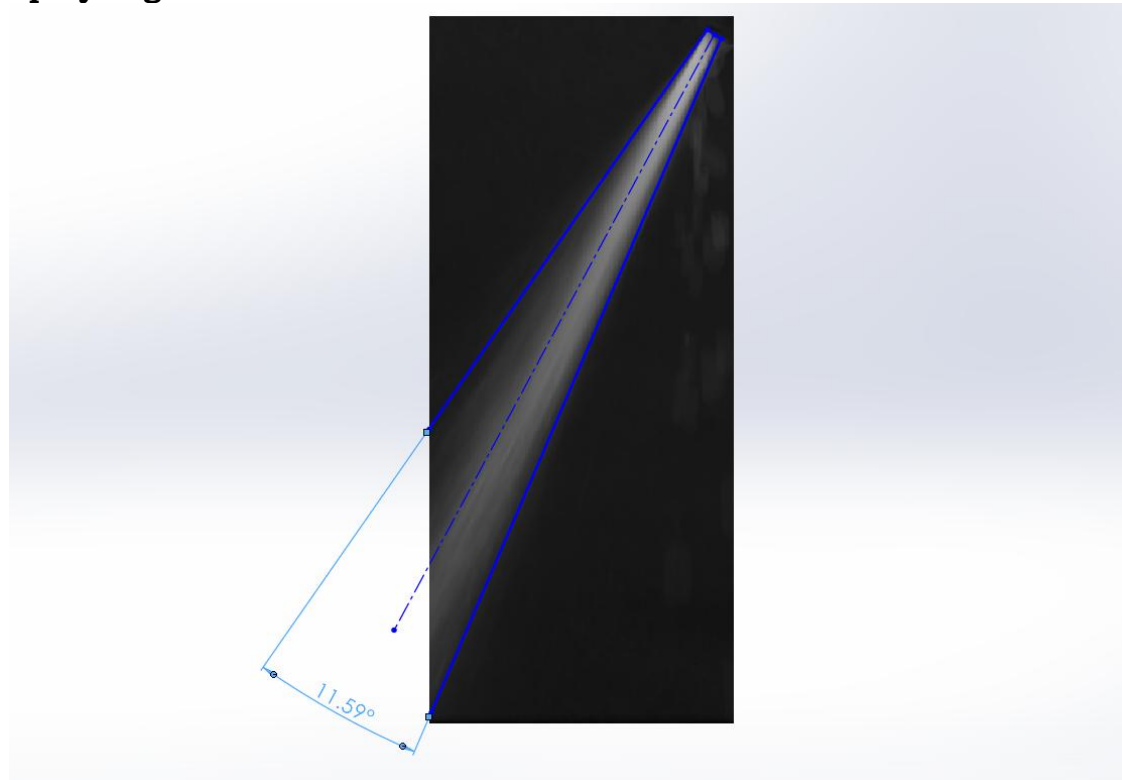


Figure 5.1.3) Spray Angle

5.2) TRIAL 2:



Density distribution:

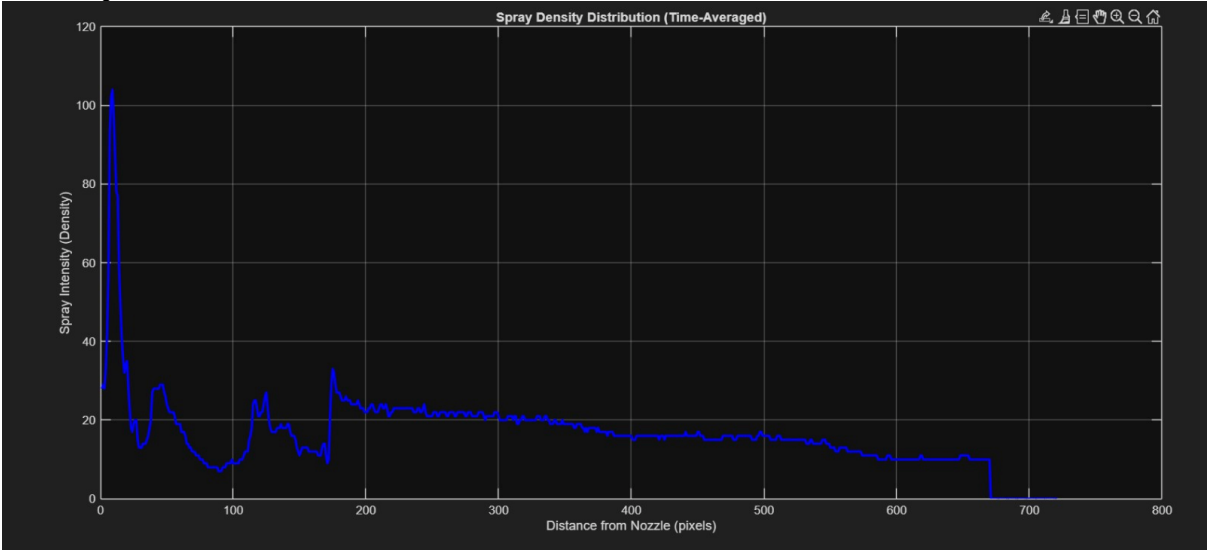


Figure 5.2.1) Density Distribution

Breakup Length: 285.68 mm
Penetration Length: 512.50 mm

Intensity Distribution:

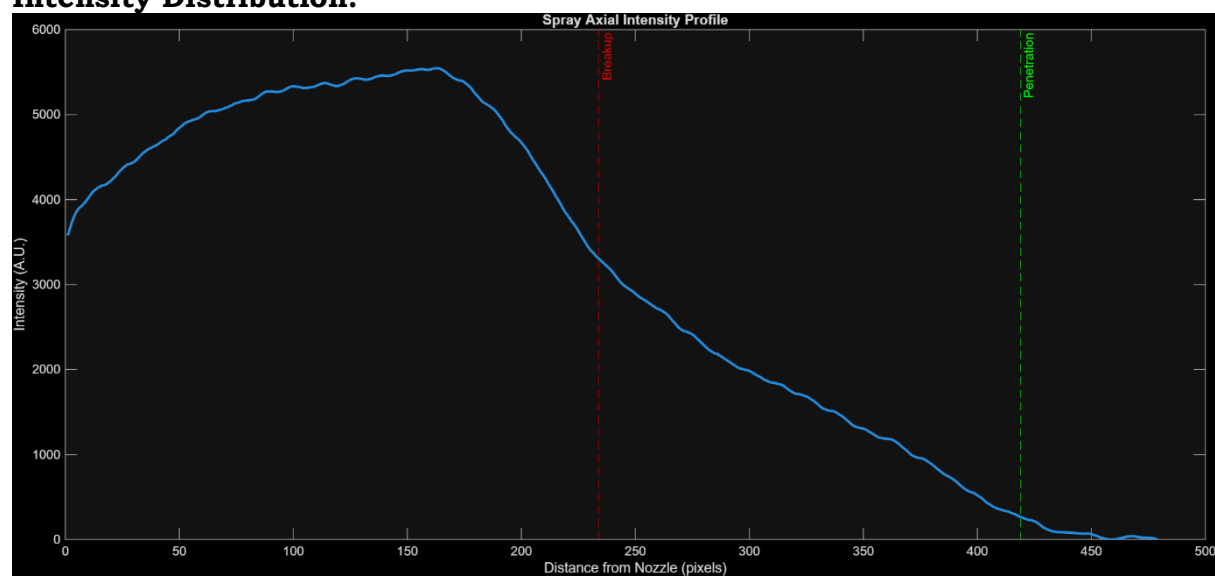


Figure 5.2.2) Intensity Distribution

Spray angle on SolidWorks:

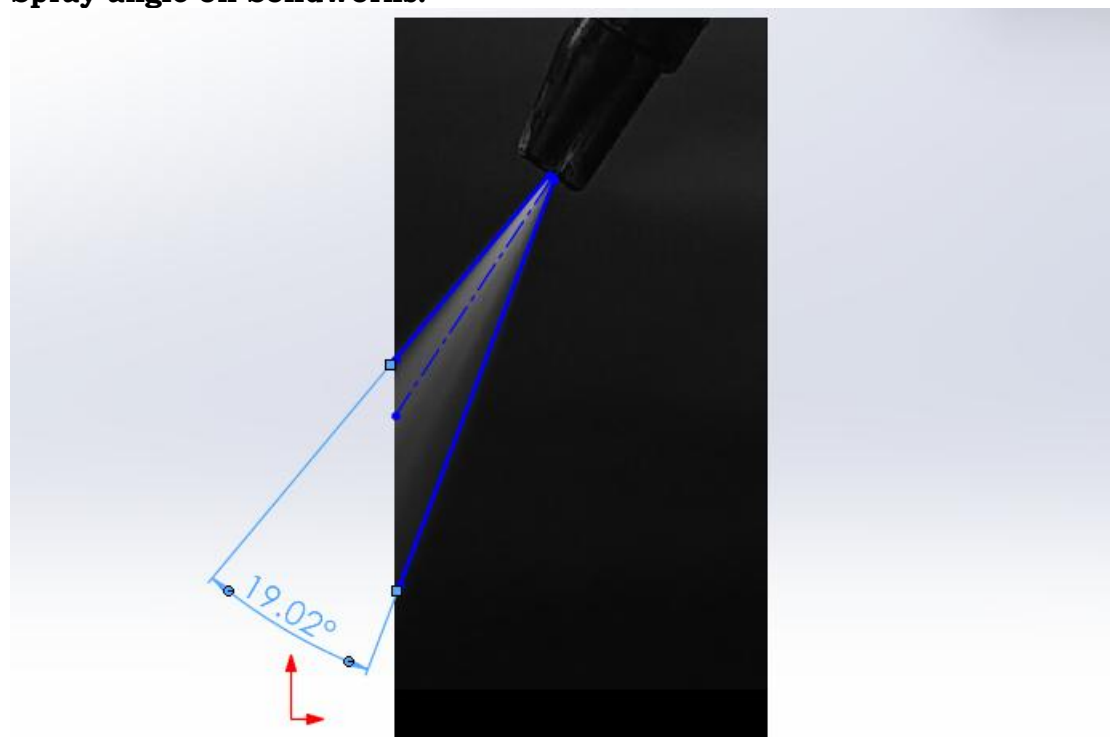


Figure 5.2.3) Spray Angle

5.3) TRIAL 3:



Density distribution:

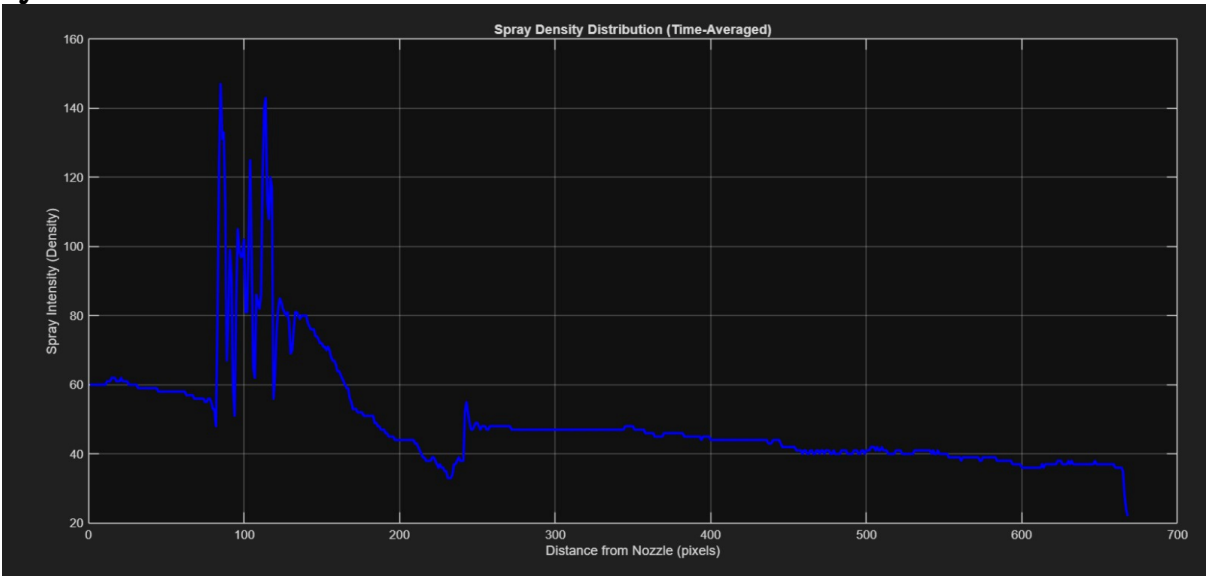


Figure 5.3.1) Density Distribution

Breakup Length: 308.92 mm
Penetration Length: 309.66 mm

Intensity profile:

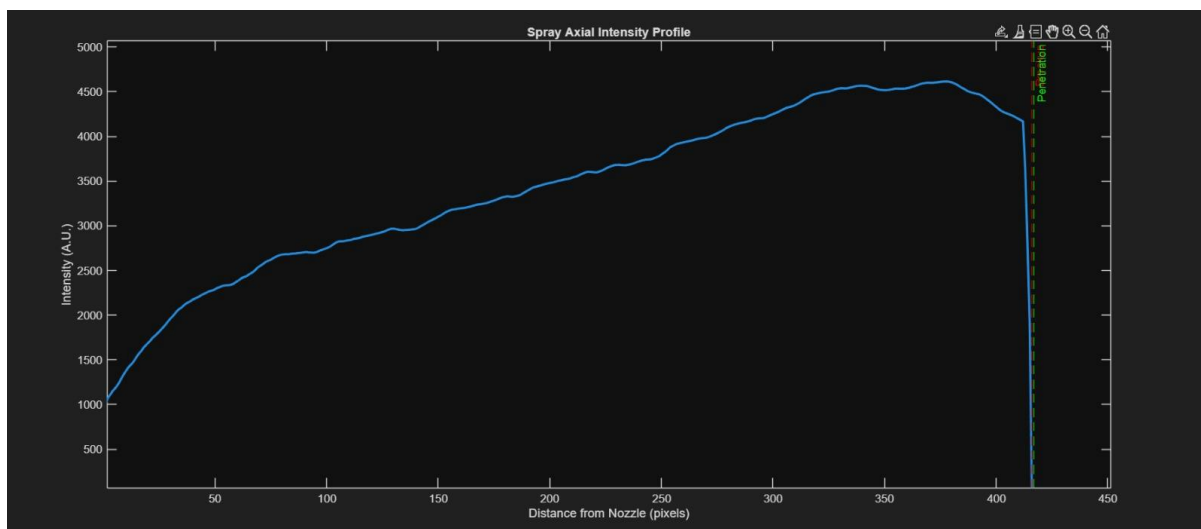


Figure 5.3.2) Intensity Distribution

Spray angle on SolidWorks:

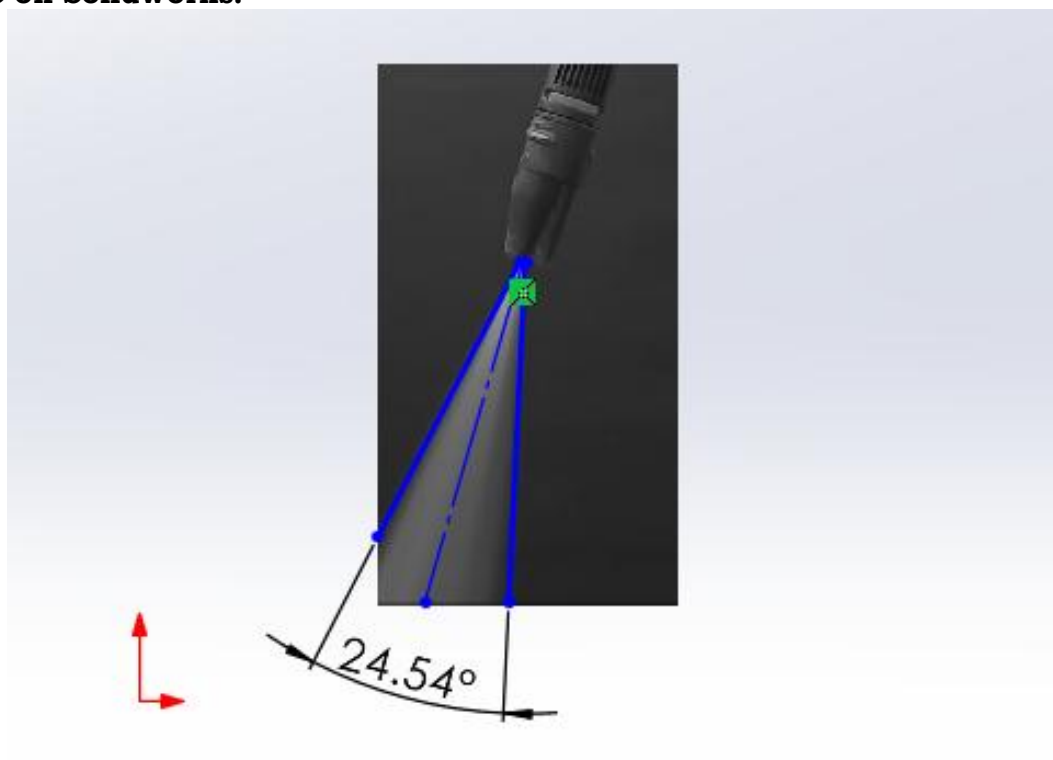


Figure 5.3.3) Spray Angle

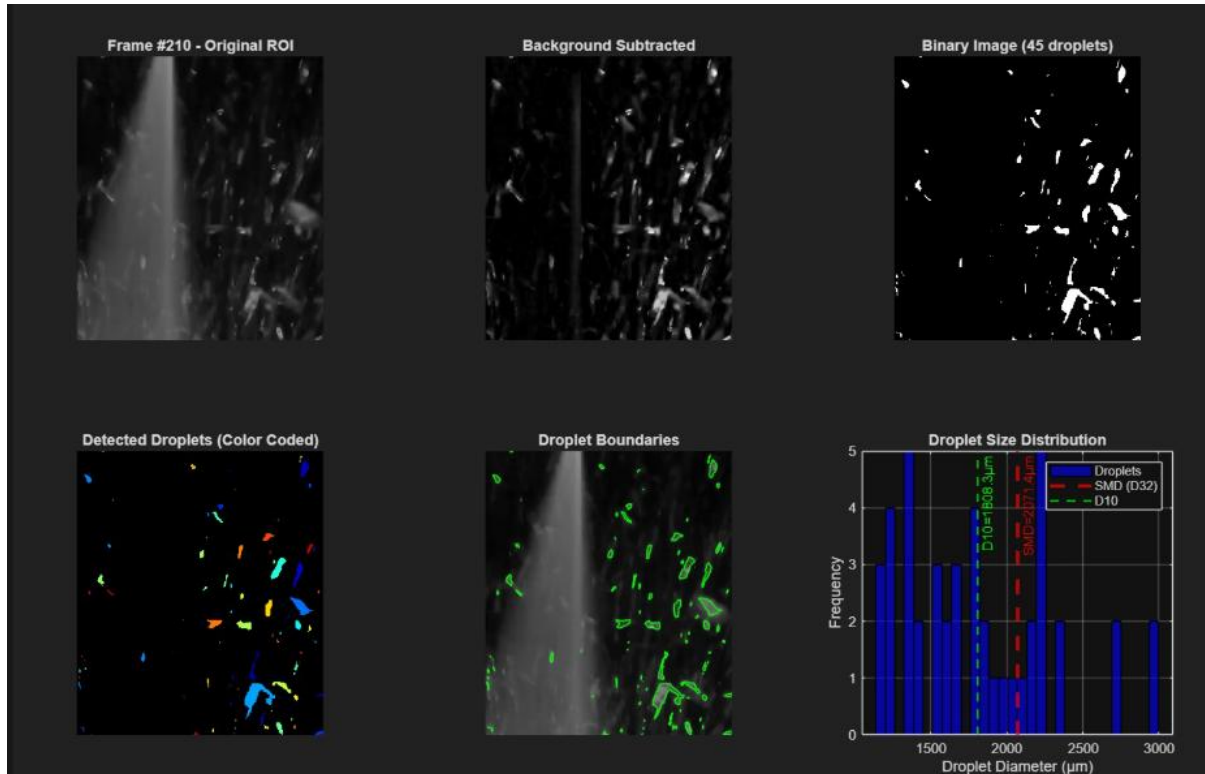


Figure 5.3.4: Sauter Mean Diameter

D10 (Arithmetic mean): 1808.27 μm
 D20 (Surface mean): 1871.95 μm
 D30 (Volume mean): 1936.20 μm
 SMD/D32 (Sauter mean): 2071.40 μm
 Min diameter: 1142.53 μm
 Max diameter: 2979.35 μm

NOZZLE WIDTH (mm)	BREAKUP LENGTH (mm)	PENETRATION LENGTH (mm)
3	180.19	227.71
4	285.68	512.50
5	308.92	309.66

Table 5: Results

The experimental data reveals a clear correlation between nozzle diameter and the spatial reach of the spray. Trial 1 (3mm nozzle, 0.105 kg/s) exhibited a Breakup Length of 180.19 mm and a Penetration Length of 227.71 mm. This suggests that at lower mass flow rates and smaller orifice sizes, the liquid core is more susceptible to aerodynamic instabilities, leading to earlier fragmentation. In contrast, Trial 2 (4mm nozzle, 0.145 kg/s) demonstrated a significant increase in both metrics, with a Breakup Length of 285.68 mm (+58.5%) and a Penetration Length of 512.50 mm (+125%). This substantial jump in penetration indicates

that the higher mass flow rate imparts greater axial momentum to the jet, allowing it to penetrate deeper into the ambient air before being disrupted by Kelvin-Helmholtz instabilities.

The Intensity and Density Profiles (Images 11, 12, 15, and 16) provide a visual map of these transitions. In Trial 1, the intensity decay is more gradual, whereas Trial 2 shows a more robust core followed by a sharp drop at the breakup point, indicating a more concentrated mass flux in the larger nozzle setup. For Trial 3 (5mm nozzle, 0.145 kg/s), while the mass flow rate remained equal to Trial 2, the larger orifice area likely reduced the exit velocity, which typically leads to a further extension of the breakup length as the Weber number (the ratio of disruptive aerodynamic force to restorative surface tension) decreases at lower velocities.

The Sauter Mean Diameter (SMD) analysis is the final critical metric for propulsive performance. While Trial 2 provides the greatest spatial reach (penetration), it likely produces larger initial droplets compared to Trial 1. In a combustion environment, the goal is to maximize penetration while minimizing SMD to ensure the fuel reaches the center of the chamber but evaporates quickly. The Spray Cone Angle measurements from SolidWorks complete the picture; a wider angle in Trial 3 would suggest better macro-mixing but must be weighed against the risk of wall impingement, which Trial 2's high penetration already highlights as a design constraint. Collectively, these three trials validate that the low-cost Bosch system effectively mimics the complex atomization regimes required for aerospace injector testing.

